Biology Class 11 National Book Foundation FBISE Quality Notes
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Chapter 15: Ecology

Ecology: *The study of the relationships of living organisms with each other and with their physical surroundings. It is a way to study the structure and function of nature.*

15.1 Ecosystem (SLO 1)

- **Ecosystem:** *the major unit of ecology; a self-sustained geographical area where living organisms interact with their non-living environment (atmosphere, hydrosphere, soil and biological communities). Size from a small pond to a rainforest or the whole planet.*
- **Biodiversity:** *variety of species of plants, animals and microorganisms in an ecosystem; essential for stability and resilience.*

Term	Meaning
Species	Group of related organisms that interbreed freely in nature and produce fertile offspring of their kind.
Population	Members of a species living in a geographical area at a given time.
Community	Interacting populations of different species (competition, predation, symbiosis).
Ecosystem	Community + its non-living environment.
Biosphere	The thin life-containing blanket around the earth; includes all ecosystems.

- **Conditions for life on earth:** *steady supply of solar energy, water as the medium of life, suitable temperature range, correct proportion of minerals, and screening of ultraviolet radiation by ozone.*

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Components of an ecosystem (SLO 2, 3)

- **Abiotic components:** non-living factors: light, air, water, soil, basic elements and compounds. **Biotic components:** producers, consumers and decomposers.

Biotic group	Details and examples
Producers (autotrophs)	Plants, algae and photosynthetic bacteria make complex organic compounds from inorganic raw materials; basis of any ecosystem. On land mainly plants; in water **phytoplankton** and shallow-water rooted plants.
Primary consumers (herbivores)	Feed directly on producers: cattle, deer, rabbit, grasshopper; aquatic: crustaceans, molluscs, herbivorous fish.
Secondary consumers	Carnivores feeding on primary consumers: many fishes, frogs, snakes, predatory birds, lion, tiger.
Tertiary consumers	Feed on secondary consumers: wolf, owl, hawk.
Decomposers	Bacteria and fungi secrete digestive enzymes on dead matter and wastes, absorb some products and release the rest to the environment: recycle materials for producers.

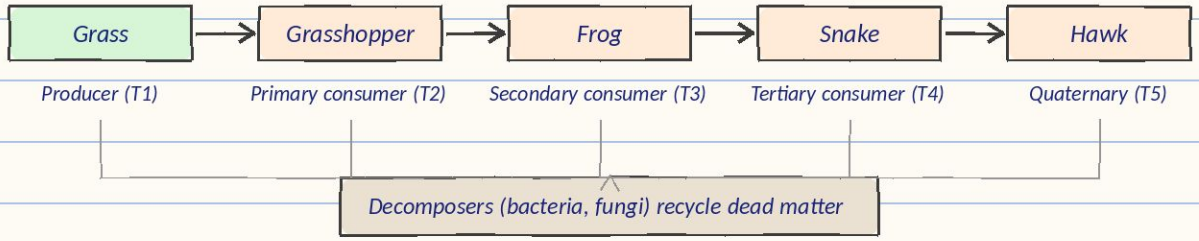
Modes of nutrition: plants as major autotrophs (SLO 3)

- Plants convert **light energy to chemical energy** in photosynthesis: CO_2 enters through stomata to mesophyll cells; water absorbed by roots rises in xylem by transpiration pull; in mesophyll CO_2 and water make glucose using light trapped by chlorophylls and other pigments. **Consumers** are heterotrophs (animals, fungi, protozoans, many bacteria) that cannot make their own food.

Food chain, food web and trophic levels (SLO 4)

Food chain: The sequence of organisms through which food and nutrients move in an ecosystem.

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Energy flows one way (arrows point to the eater); nutrients are recycled

Food web: interconnected food chains give alternatives at each trophic level (grass: grasshopper, mouse; mouse: snake, hawk; frog: snake, hawk)

Fig: A grassland food chain and trophic levels

- *Food chains rarely occur alone; interconnected chains form a **food web** (grass is eaten by grasshopper and mouse; mouse by snake or hawk; grasshopper by frog, frog by snake and hawk). A web offers **food alternatives** at each trophic level and helps keep the ecosystem balanced.*

***Trophic level:** Each stage of the food chain: a position in the food chain showing the source of energy and nutrients.*

- *T1 producers; T2 primary consumers (herbivores); T3 secondary consumers (carnivores eating herbivores); T4 tertiary consumers (carnivores eating carnivores); T5 quaternary consumers. Fewer organisms at higher levels because less energy is available.*

Productivity (SLO 17)

- *Sun is the ultimate energy source; energy flows through consumers **one way (non-cyclic)**. **Productivity:** rate at which producers create organic matter by photosynthesis, measured as biomass per unit area per unit time.*
- ***Gross primary productivity (GPP):** total solar energy fixed by producers in photosynthesis; depends on light, water, nutrients and temperature. **Net primary productivity (NPP):** energy left after producers meet their respiratory needs (NPP = GPP - respiration); appears as producer biomass; it supports the entire food web (herbivores, carnivores, decomposers).*

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15.2 Energy flow (SLO 5, 16)

- Even under optimum conditions only about **1%** of radiation is converted into GPP. Energy flow obeys the laws of thermodynamics. **Loss between trophic levels** (about 90% at each level, 10% passes on) occurs by respiration, growth, reproduction, non-predatory death (organisms that die but are not eaten) and heat, and as excretory products or faeces. So total energy decreases progressively along the chain.

Ecological pyramids (SLO 16)

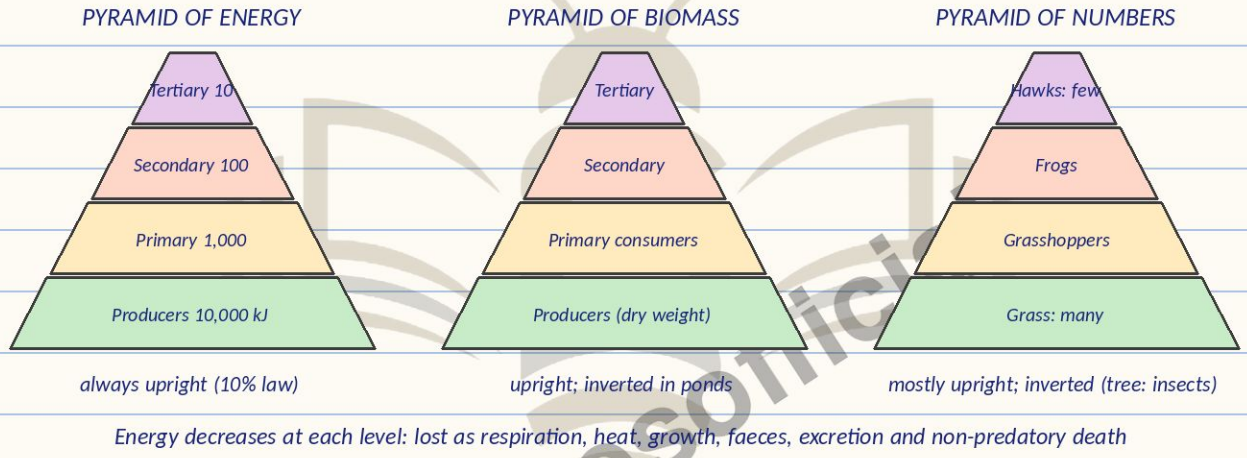


Fig: Ecological pyramids

- **Pyramid of energy:** the most useful; energy at each level; producers at the base have the most energy (they fix light energy) and energy falls regularly toward the top because some is used for life activities and lost as waste or heat. **Pyramid of biomass:** total dry weight of living material at each level; largest at producers, decreasing upward. **Pyramid of numbers:** number of organisms at each level; not always regular, but mostly decreases, with many producers and few top consumers.

15.3 Biogeochemical cycles (SLO 9)

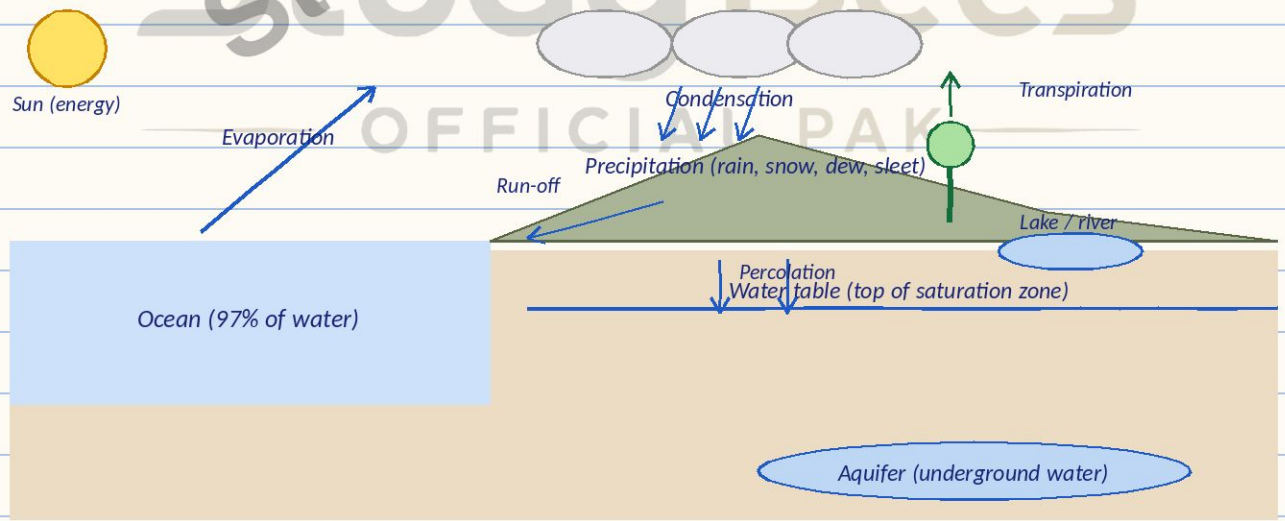
Biogeochemical cycle: The cyclic flow of nutrients from the environment to organisms and back to the environment.

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- ***Primary reservoirs of nutrients:*** *each reservoir is defined by organic or inorganic material, and whether it is available to organisms. (a)* ***Living organisms and detritus:*** *organic, available (consumers and detritivores feed on it); (b)* ***fossilized organic matter:*** *coal, oil, peat; not available until burned; (c)* ***inorganic nutrients dissolved in soil water and air:*** *available; returned by respiration, excretion and decomposition; (d)* ***rocks:*** *inorganic, unavailable except slowly by weathering and erosion.*
- ***Role of decomposers in nutrient cycling:*** *break dead matter into simple molecules (ammonia, CO_2, minerals), returning them to the available inorganic reservoir for producers.*

Water cycle (SLO 10, 11)

- *Ocean is the major reservoir (over 97% of water). The cycle is driven by* ***solar energy*** *(evaporation) and* ***gravity*** *(precipitation: rain, snow, dew, sleet). Water falling on land evaporates from soil, lakes and streams; some runs off to the oceans; some* ***percolates*** *into the ground and saturates it. The top of the saturation zone is the* ***water table****; where basins or channels reach it, water appears as* ***lakes and ponds*** *(basins) or* ***streams and rivers*** *(channels). Water in underground rivers or layers is called an* ***aquifer****.*



Driven by solar energy (evaporation) and gravity (precipitation, run-off)

Fig: Water cycle

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- ***Water in living bodies:*** *bodies are about 70% water; roots absorb it and most evaporates from leaves (transpiration); a little combines with CO_2 in photosynthesis and is released in respiration; heterotrophs get water from food and drinking.*

Nitrogen cycle (SLO 12)

- *Nitrogen is needed for amino acids, nucleic acids and proteins. The cycle is continuous, maintained by decomposers and other bacteria:* ***ammonification, nitrification, nitrogen fixation, assimilation, denitrification.***

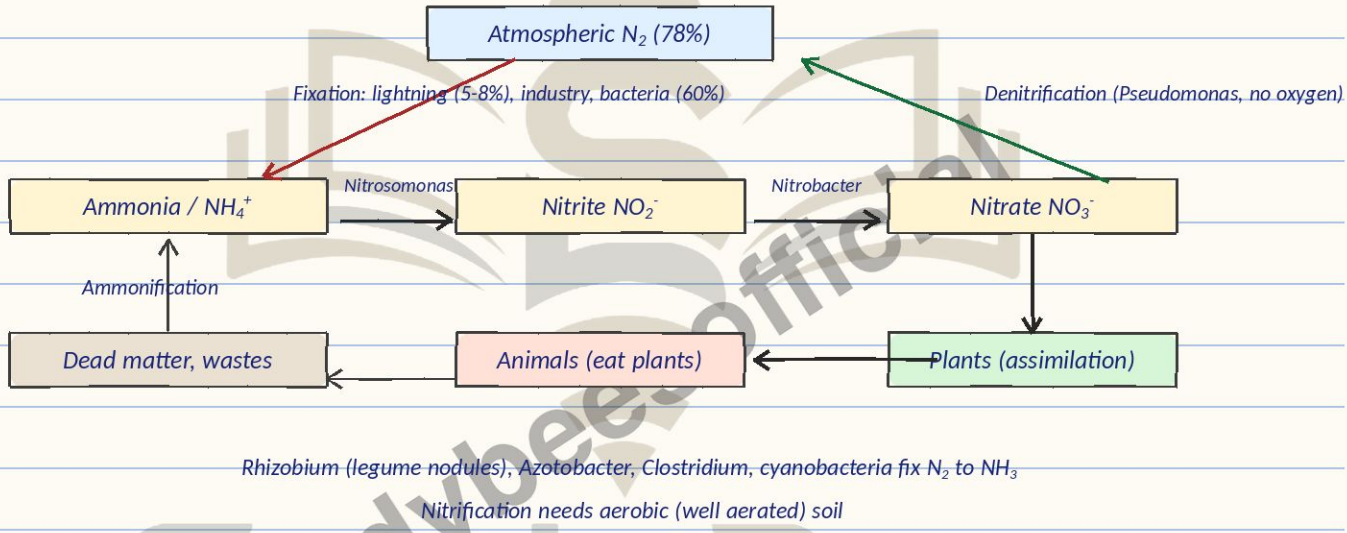


Fig: Nitrogen cycle

Process	***Description***
Ammonification	*Saprophytic soil bacteria and fungi decompose wastes and dead organisms to amino acids, then* ***ammonia / ammonium****; in aerobic soil.*
Nitrification	*Ammonia to nitrites by* ***Nitrosomonas****; nitrites to nitrates by* ***Nitrobacter****; only in well aerated soil (aerobic bacteria).*

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Process	Description
Nitrogen fixation	N_2 has a very strong triple bond. **Atmospheric:** lightning makes oxides of nitrogen carried by rain (5-8%). **Industrial:** nitrogen fertilizers. **Biological:** bacteria fix about 60%: $N_2 + 3H_2$ gives $2NH_3$; free-living **Azotobacter** (aerobic), **Clostridium** (anaerobic) and symbiotic **Rhizobium** (legumes).
Assimilation	Use of nitrogen compounds in living bodies: plants take up ammonia and nitrates; animals eat plants or other animals; many microbes use free N_2.
Denitrification	In the absence of oxygen, **Pseudomonas** and other bacteria break nitrates (NO_3 to NO_2 to NO / N_2O to N_2) and release nitrogen gas to the air.

15.4 Ecological succession (SLO 13-15)

Ecological succession: The process by which species are replaced over time in an area: the orderly, predictable change in community structure from simple to complex. Early species are replaced because later species are better able to grow and reproduce under the changed conditions.

- **Sere:** an individual succession; **seral stages:** its developmental phases; **seral communities:** groups at each stage; **pioneers:** first migrants that take hold of the bare area; **climax community:** the end product, a relatively stable community.

Primary succession	Secondary succession
In a habitat not previously inhabited (bare rock, new lava, rock scraped by glaciers)	In a habitat already substantially modified by a previous community; soil already present
Starts without soil; takes very long	Faster: abandoned farm fields reverting to forest; forest cleared by fire, flood, cyclone
Pioneers: lichens	Pioneers: grasses, weeds

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Hydrarch succession	Xerarch succession
Begins in water (ponds, lakes, marshes)	Begins in a dry habitat with water deficiency (bare rocks, sand dunes, rocky slopes)
Stages called hydroseres	Stages called xeroseres
Ends in a mesophytic forest	Ends in a mesophytic forest (climax)

Xerarch succession on bare rock (SLO 15)

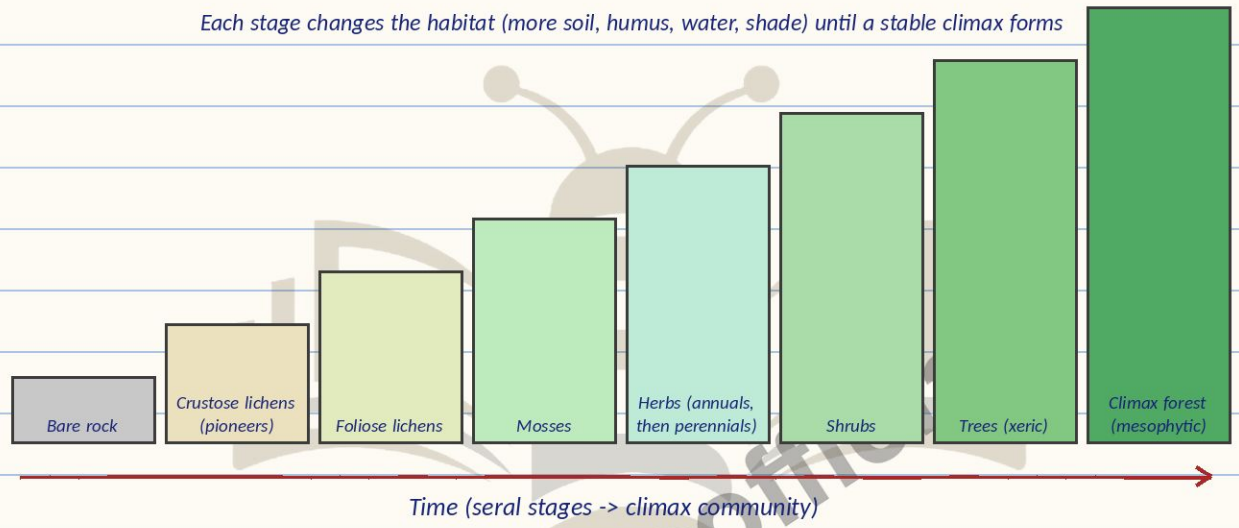


Fig: Xerarch (primary) succession on bare rock

1. **Crustose lichen stage (pioneer):** crust-like lichens (Lecanora, Rhizocarpon) tolerate desiccation, absorb rain like sponges and decompose rock by secreting acids.
2. **Foliose lichen stage:** leaf-like lichens (Parmelia, Dermatocarpon) on the little soil; collect a thin layer of soil.
3. **Moss stage:** xerophytic mosses in crevices; increase soil; death and decay form a mat that holds more water.
4. **Herb stage:** annual weeds, roots penetrate deep and secrete acids (more weathering), leaf litter adds humus; shading reduces evaporation; then biennial and perennial herbs and xeric grasses.
5. **Shrub stage:** xeric shrubs (from seeds or rhizomes) overshadow herbs; birds disperse seeds; dense growth.
6. **Tree stage:** tree saplings grow among shrubs; canopy shades the area;

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shade-loving plants form secondary vegetation; litter and roots deepen the soil.

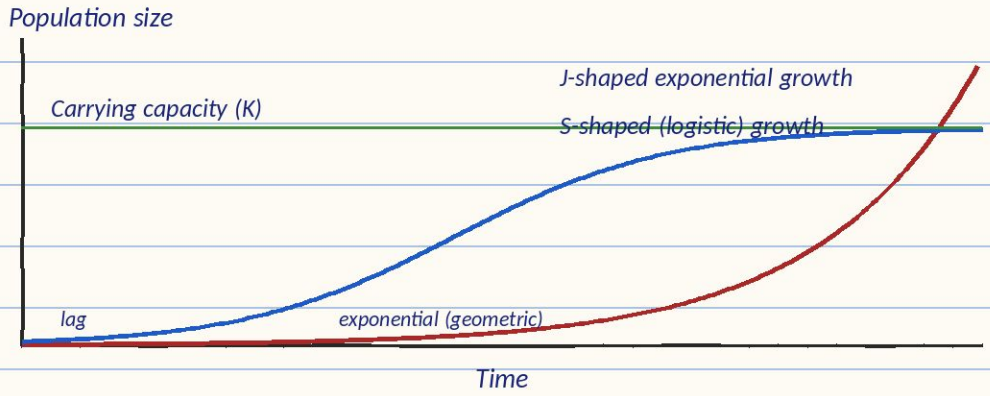
7. ***Climax stage:*** *the first (xeric) trees are replaced by* ***mesophytic*** *trees as soil deepens: a forest develops.*

15.5 Population dynamics (SLO 18, 19)

- *Population dynamics = long- and short-term changes in population size and the factors regulating it.* ***Inflow:*** *births, immigration.* ***Outflow:*** *culling, predation, natural deaths, accidents, emigration.*
- ***Population:*** *group of individuals of a single species in the same general area, relying on the same resources and likely to interact and breed. Characteristics: growth, density, distribution, carrying capacity, viable size.*

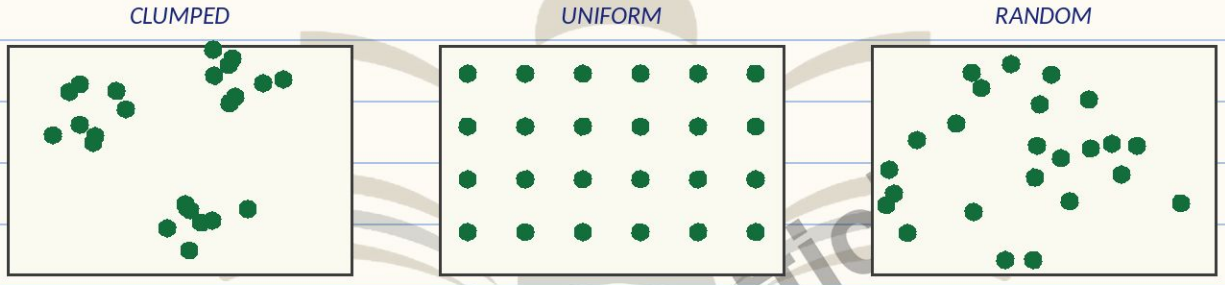
Characteristic	***Meaning***
Growth	*Increase in number of individuals. A few individuals in an unoccupied area: lag phase, then* ***exponential growth*** *(doubles at regular intervals; geometric growth).*
Density	*Number of individuals per unit area or volume (mulberry trees per km^2; E. coli per ml).*
Distribution	*Dispersion of individuals:* ***clumped*** *(patches),* ***uniform*** *(evenly spaced),* ***random****.*
Carrying capacity	*Maximum population size an environment can support; varies with space, time and limiting resources.*
Minimum viable size	*Smallest number of individuals needed for the population to persist (avoids extinction by inbreeding and chance events).*

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Lag phase, exponential phase (doubles at regular intervals), then growth levels off at K

Fig: Population growth curves



Clumped: patches (resources unevenly spread); Uniform: even spacing (competition); Random: unpredictable

Fig: Distribution patterns in a population

Human impact on ecosystems (SLO 19)

Impact	Examples
Habitat destruction and fragmentation	Deforestation, urbanization, roads and dams.
Pollution	Air (acid rain, smog), water, soil pollution.
Climate change	Greenhouse gas emissions, temperature changes, sea level rise.
Overexploitation	Overfishing, hunting and poaching, defaunation.
Invasive species	Non-native species outcompete native ones; change soil, nutrients and fire.
Land use changes	Agricultural expansion, monoculture farming.
Loss of biodiversity	Extinction and genetic erosion.

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Impact	Examples
Resource depletion	Depleted aquifers, soil erosion and desertification.

- **Role of the Population Welfare Department (Pakistan):** family planning and reproductive health services through Family Welfare Centres, Mobile Service Units and Reproductive Health Services centres.

15.5.3 The greenhouse effect (SLO 6, 7)

- A **natural** process that warms the Earth by trapping heat radiation: sunlight warms the surface and the rest is radiated back; **greenhouse gases** (GHGs) absorb part of the outgoing heat and keep the surface warmer, making Earth a comfortable place. Human activity (fossil fuels, deforestation) has increased GHG levels, enhancing the effect and causing **global warming**.

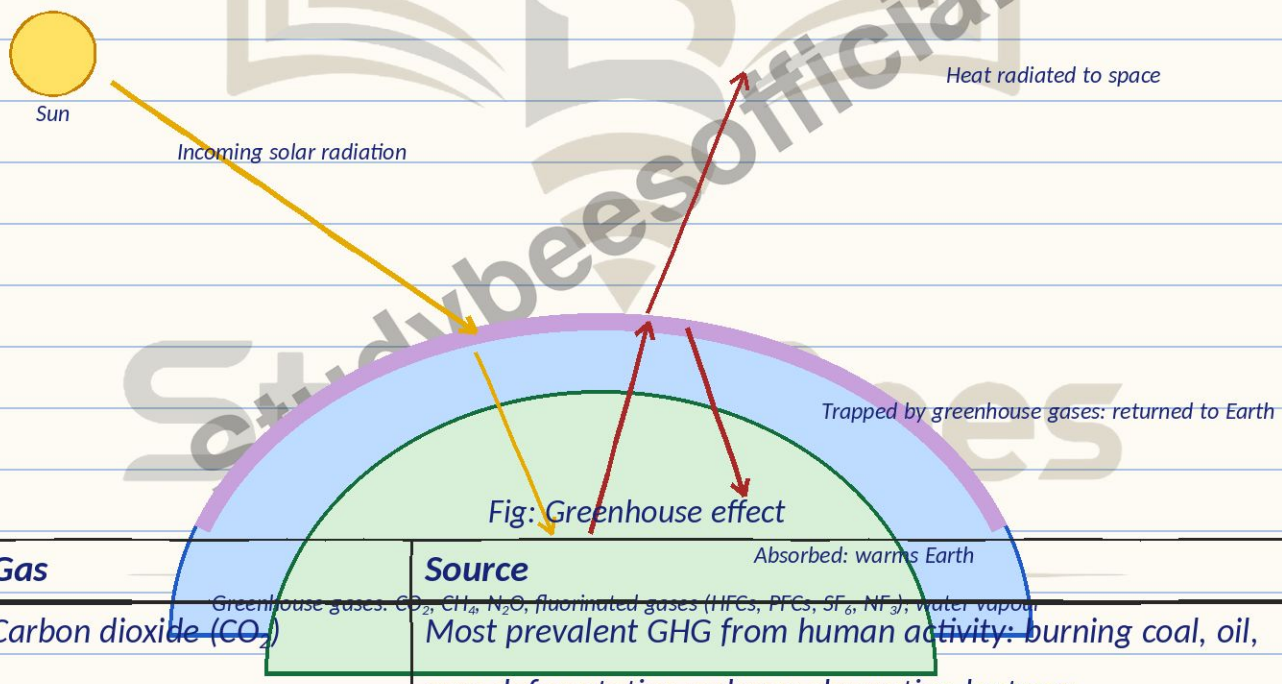


Fig: Greenhouse effect

Gas	Source
Carbon dioxide (CO_2)	Most prevalent GHG from human activity: burning coal, oil, gas; deforestation reduces absorption by trees.
Methane (CH_4)	Much more potent heat trapper than CO_2 but less concentrated: coal, oil and gas production and transport; livestock and agriculture.
Nitrous oxide (N_2O)	Agriculture, industry, combustion of fossil fuels and solid waste.

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Gas	Source
Fluorinated gases (HFCs, PFCs, SF_6, NF_3)	Synthetic industrial gases released during production and use.

Harmful effects of greenhouse gases (SLO 7)

1. **Global warming:** long-term rise in average surface temperature; heat waves, changed weather patterns.
2. **Climate change:** altered rainfall, intense storms, droughts, floods.
3. **Melting polar ice and glaciers:** sea level rise threatens coasts; habitats of polar bears and penguins lost.
4. **Ocean acidification:** excess CO_2 lowers sea pH, harming corals and shellfish.
5. **Biodiversity loss:** species shift or go extinct.
6. **Impact on agriculture:** reduced yields from droughts or heat; pests; food shortages and prices.
7. **Health impacts:** heat illnesses, respiratory problems, insect-borne diseases (mosquitoes).
8. **Economic consequences:** disaster response, infrastructure damage, healthcare costs, lower productivity.

- Remedies: reduce emissions, shift to renewable energy, adapt to climate change.

15.6 Coral reefs (SLO 8)

- Diverse, productive marine ecosystems in shallow tropical and subtropical waters; colonies of **coral polyps** secrete calcium carbonate skeletons that form the reef ("**rainforests of the sea**"). Corals have a **symbiotic relationship with photosynthetic algae** that feed them and give colour. Reefs host thousands of species (fish, predators, herbivores, sea grasses, algae) and act as **natural barriers** protecting coasts from erosion and storms.

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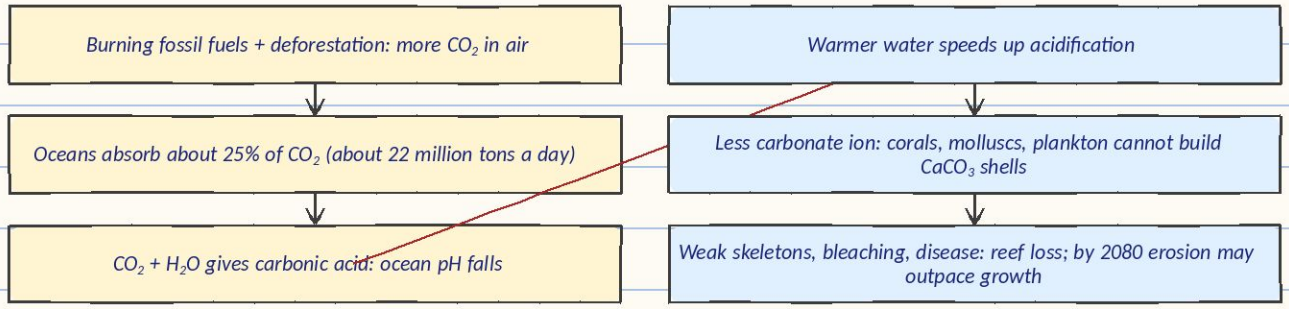


Fig: Ocean acidification and coral reefs

- ***Ocean acidification:*** *oceans absorb about 25% of CO_2 released (about 22 million tons per day); CO_2 + water gives carbonic acid and lowers pH; warming speeds it up. Calcifying organisms (corals, molluscs, some plankton) find it harder to make shells and skeletons: weaker structures, predation and disease.* ***More than half of the world's corals are under threat;*** *by 2080 coral erosion may exceed rebuilding.*
- ***Corals as barometer of aquatic ecosystem health:*** *corals are very sensitive to changes in pH, temperature and CO_2; bleaching, slowed growth and reef loss are early warnings of ocean acidification and warming, so coral condition indicates the health of the whole marine ecosystem.*

15.7 Ecosystems of Pakistan (SLO 21)

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Ecosystem	Climate, flora, fauna, productivity, threats
Grassland	Tropical (savanna; monsoon rain 200-600 mm; above 40 C in summer; moderate to high productivity) and temperate (rain 150-500 mm; summers 25-35 C, cold winters; Balochistan, parts of Punjab; lower productivity). Flora: Cenchrus ciliaris, Themeda anathera, Apluda mutica, Panicum antidotale, Dicanthium, Stipagrostis, Heteropogon. Fauna: deer, antelopes, gazelles, wild goats, rodents, birds; predators: fox, wolf, jackal, big cats, birds of prey. Threats: overgrazing, soil erosion, agriculture, urbanization, climate change; conservation by sustainable land management.
Desert	Thar, Tholistan (Cholistan), Thal, Kharan and the cold desert of Skardu. Rain 100-250 mm (July-September); summer above 40 C (up to 50 C), winter below 10 C. Flora (xerophytes): cacti, Acacia, Prosopis, succulents. Fauna: jerboas, gerbils, desert fox, wildcats, snakes, lizards, sandgrouse, larks, raptors, ants, beetles. Low primary productivity; adaptations: deep roots, concentrated urine, nocturnal habit, burrowing; scavengers recycle nutrients.
Coniferous alpine and boreal forests	Western Himalaya and Karakoram. Cold; moderate to high precipitation (snow in winter, rain in summer). Flora: pine, spruce, fir, cedar (needle leaves), rhododendron, juniper. Fauna: snow leopard, brown bear, Himalayan tahr, red fox, deer, golden eagle, griffon vulture, chukar. Slow productivity (short growing season, poor soil) but biodiversity hotspots.
Temperate deciduous forests	Northern Pakistan, Azad Kashmir, Himalaya and Karakoram foothills. Moderate rain through the year, warm summers, cool winters. Flora: oak, maple, pine, ash, walnut, deodar; ferns, mosses, wildflowers. Fauna: pheasants, partridges, eagles, Himalayan brown bear, snow leopard, deer, monkeys. Productivity depends on soil, climate, plant diversity; they regulate water flow and fertility; reduced by deforestation and unsustainable logging.

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Chapter 15: Short Questions with Answers (SLO based)

Q1. [SLO 1] Define ecology and ecosystem.

Ans: Ecology: study of relationships of living organisms with each other and their physical surroundings. Ecosystem: self-sustained area where living organisms interact with their non-living environment.

Q2. [SLO 1] Define species, population, community.

Ans: Species: interbreeding group producing fertile offspring. Population: members of a species in an area at a given time. Community: interacting populations of different species.

Q3. [SLO 1] Explain the term biosphere.

Ans: The thin life-containing blanket around the earth that includes all ecosystems.

Q4. [SLO 1] What conditions make life on earth possible?

Ans: Steady solar energy, water as medium of life, suitable temperature range, proper minerals and ozone screening of UV.

Q5. [SLO 1] What is biodiversity? Why is it important?

Ans: Variety of species in an ecosystem; essential for stability and resilience.

Q6. [SLO 2] What are abiotic and biotic components?

Ans: Abiotic: non-living (light, air, water, soil, elements). Biotic: living (producers, consumers, decomposers).

Q7. [SLO 2] What are producers, consumers and decomposers?

Ans: Producers: autotrophs making food. Consumers: heterotrophs depending on producers. Decomposers: bacteria and fungi breaking down dead matter.

Q8. [SLO 2] Differentiate primary, secondary and tertiary consumers.

Ans: Primary feed on producers (herbivores); secondary feed on primary consumers; tertiary feed on secondary consumers.

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Q9. [SLO 2] What are phytoplankton?

Ans: Floating photosynthetic organisms (mainly algae), the main producers of aquatic ecosystems.

Q10. [SLO 3] How are plants major autotrophs?

Ans: They convert light energy to chemical energy in photosynthesis using CO_2 and water, forming the base of terrestrial ecosystems.

Q11. [SLO 9] Explain the role of decomposers in the nutrient cycle.

Ans: They secrete enzymes on dead matter, release simple substances such as ammonia and CO_2, and return nutrients to the environment for producers.

Q12. [SLO 4] Define food chain and food web.

Ans: Food chain: sequence of organisms through which food and nutrients move. Food web: interconnected food chains.

Q13. [SLO 4] Why is a food web important?

Ans: It gives food alternatives at each trophic level and maintains balance in the ecosystem.

Q14. [SLO 4] Define trophic level.

Ans: Each stage of the food chain, showing the position and source of energy of organisms.

Q15. [SLO 17] What is productivity?

Ans: Rate at which producers create organic matter by photosynthesis, measured as biomass per unit area per unit time.

Q16. [SLO 17] Differentiate GPP and NPP.

Ans: GPP: total solar energy fixed by producers. NPP: energy left after the producers' respiration (GPP - respiration).

Q17. [SLO 17] Why is NPP an important ecological parameter?

Ans: It is the energy available to support the whole food web.

Q18. [SLO 17] Describe productivity in desert ecosystem.

Ans: Low primary productivity because of scarce water and limited plant cover; drought-resistant grasses and shrubs form the food chain base.

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Q19. [SLO 5] What percentage of solar energy is converted into GPP?

Ans: Only about 1% under optimum conditions.

Q20. [SLO 5] How is energy lost between trophic levels?

Ans: By respiration, growth, reproduction, non-predatory death, heat, excretory products and faeces; about 90% is lost at each level.

Q21. [SLO 16] Name three ecological pyramids.

Ans: Pyramid of energy, of biomass and of numbers.

Q22. [SLO 16] Explain the pyramid of energy.

Ans: Shows energy at each trophic level; largest at producers and decreasing regularly upward because energy is used in life activities and lost as heat and waste.

Q23. [SLO 16] Why is the pyramid of energy always upright?

Ans: Energy decreases at each level due to loss, so the base always has the most.

Q24. [SLO 16] Define biomass.

Ans: The dry weight of all living material in the ecosystem.

Q25. [SLO 6] What is the greenhouse effect?

Ans: Natural warming of earth because gases trap outgoing heat radiation.

Q26. [SLO 6] Name greenhouse gases.

Ans: Carbon dioxide, methane, nitrous oxide and fluorinated gases (also water vapour).

Q27. [SLO 6] Which greenhouse gas traps heat better, CO_2 or methane?

Ans: Methane has much higher heat-trapping ability but is present in lower concentration.

Q28. [SLO 7] List harmful effects of greenhouse gases.

Ans: Global warming, climate change, melting ice and sea level rise, ocean acidification, biodiversity loss, reduced crop yield, health risks and economic loss.

Q29. [SLO 8] Why are coral reefs called rainforests of the sea?

Ans: Because of their incredible biodiversity and complex relationships.

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Q30. [SLO 8] Describe the symbiosis in corals.

Ans: Corals have a symbiotic relationship with photosynthetic algae that provide nutrients and colour.

Q31. [SLO 8] What is ocean acidification?

Ans: Fall in ocean pH because CO_2 dissolves to form carbonic acid.

Q32. [SLO 8] How much CO_2 do oceans absorb?

Ans: About 25% of that released; about 22 million tons per day.

Q33. [SLO 8] Why are coral reefs a barometer for health of an aquatic ecosystem?

Ans: Corals are very sensitive to pH and temperature change, so their bleaching and erosion signal the health of the ecosystem.

Q34. [SLO 9] Define biogeochemical cycle.

Ans: Cyclic flow of nutrients from environment to organisms and back.

Q35. [SLO 9] Name the primary reservoirs of nutrients.

Ans: Living organisms and detritus; fossil organic matter; inorganic nutrients in soil, water and air; and rocks.

Q36. [SLO 10] What is the major reservoir of water?

Ans: The ocean, with over 97% of water.

Q37. [SLO 11] Define water table and aquifer.

Ans: Water table: top of the saturation zone of ground water. Aquifer: underground river or layer of ground water.

Q38. [SLO 11] Describe water in basins.

Ans: Where basins reach the water table, water appears as lakes and ponds; in channels as streams and rivers.

Q39. [SLO 12] What are the processes of the nitrogen cycle?

Ans: Ammonification, nitrification, nitrogen fixation, assimilation and denitrification.

Q40. [SLO 12] Describe the ways of nitrogen fixation.

Ans: Atmospheric (lightning), industrial (fertilizers) and biological (nitrogen-fixing bacteria like Rhizobium, Azotobacter, Clostridium).

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Q41. [SLO 12] Name the nitrifying bacteria.

Ans: Nitrosomonas (ammonia to nitrite) and Nitrobacter (nitrite to nitrate).

Q42. [SLO 12] What is denitrification? Name a bacterium.

Ans: Conversion of nitrates back to nitrogen gas in absence of oxygen; Pseudomonas.

Q43. [SLO 12] Why does nitrification occur only in aerated soil?

Ans: The nitrifying bacteria are aerobic.

Q44. [SLO 12] What is ammonification?

Ans: Production of ammonia or ammonium compounds from decomposition of organic matter by microorganisms.

Q45. [SLO 12] What is the role of Rhizobium?

Ans: It lives in root nodules of legumes and fixes atmospheric nitrogen to ammonia.

Q46. [SLO 13] Define ecological succession, sere and climax.

Ans: Succession: replacement of species over time. Sere: an individual succession. Climax: the final stable community.

Q47. [SLO 13] What are pioneers?

Ans: The first migrants (organisms) to take hold of a bare area, e.g. crustose lichens.

Q48. [SLO 14] Differentiate primary and secondary succession.

Ans: Primary: on habitat not previously inhabited (bare rock). Secondary: in habitat already modified, with soil present (abandoned field, burnt forest).

Q49. [SLO 14] Differentiate xerarch and hydrarch succession.

Ans: Xerarch starts on dry bare rock or sand (xeroseres); hydrarch begins in water such as ponds (hydroseres).

Q50. [SLO 15] Name stages of xerarch succession.

Ans: Crustose lichen, foliose lichen, moss, herb, shrub, tree and climax.

Q51. [SLO 15] How do lichens prepare rock for succession?

Ans: They secrete acids that decompose the rock and collect a thin soil layer.

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Q52. [SLO 18] Define population dynamics.

Ans: Long- and short-term changes in population size and the factors controlling it.

Q53. [SLO 18] List characteristics of a population.

Ans: Growth, density, distribution, carrying capacity and viable size.

Q54. [SLO 18] Define population density and distribution patterns.

Ans: Density: individuals per unit area. Distribution: clumped, uniform or random.

Q55. [SLO 18] What is carrying capacity?

Ans: The maximum population size an environment can support.

Q56. [SLO 18] What is exponential growth?

Ans: Growth in which the population doubles at regular intervals (geometric) after a lag phase.

Q57. [SLO 19] Name some effects of human population growth on ecosystems.

Ans: Habitat destruction, pollution, climate change, overexploitation, invasive species, loss of biodiversity and resource depletion.

Q58. [SLO 19] What is the role of the Population Welfare Department?

Ans: It provides family planning and reproductive health services through Family Welfare Centres, Mobile Service Units and RHS centres.

Q59. [SLO 21] Name four important ecosystems of Pakistan.

Ans: Grassland, desert, coniferous alpine and boreal forests, temperate deciduous forests.

Q60. [SLO 21] Name the deserts of Pakistan.

Ans: Thar, Tholistan (Cholistan), Thal, Kharan and the cold desert of Skardu.

Q61. [SLO 21] Write adaptations of desert organisms.

Ans: Deep roots, reduced leaves, concentrated urine, nocturnal activity, burrowing.

Q62. [SLO 21] Name trees of coniferous and deciduous forests.

Ans: Conifers: pine, spruce, fir, cedar. Deciduous: oak, maple, ash, walnut, deodar.

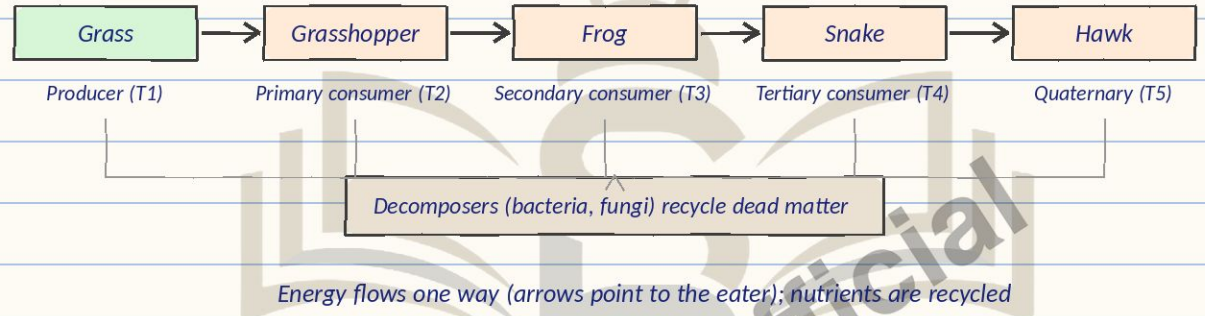
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Chapter 15: Long Questions with Answers (SLO based)

Q1. [SLO 1-4] Describe the components of an ecosystem: producers, consumers, decomposers; food chain, food web and trophic levels.

Ans:



Energy flows one way (arrows point to the eater); nutrients are recycled

Food web: interconnected food chains give alternatives at each trophic level (grass: grasshopper, mouse; mouse: snake, hawk; frog: snake, hawk)

Fig: A grassland food chain and trophic levels

- Abiotic: light, air, water, soil. Biotic: producers (T1), primary (T2), secondary (T3), tertiary (T4) consumers, decomposers. Food web: interconnected chains.

Q2. [SLO 5, 16] Explain the loss of energy between trophic levels and interpret ecological pyramids.

Ans:

- About 1% of solar radiation becomes GPP; about 90% of energy is lost at each level (respiration, growth, reproduction, non-predatory death, heat, excretory products, faeces).

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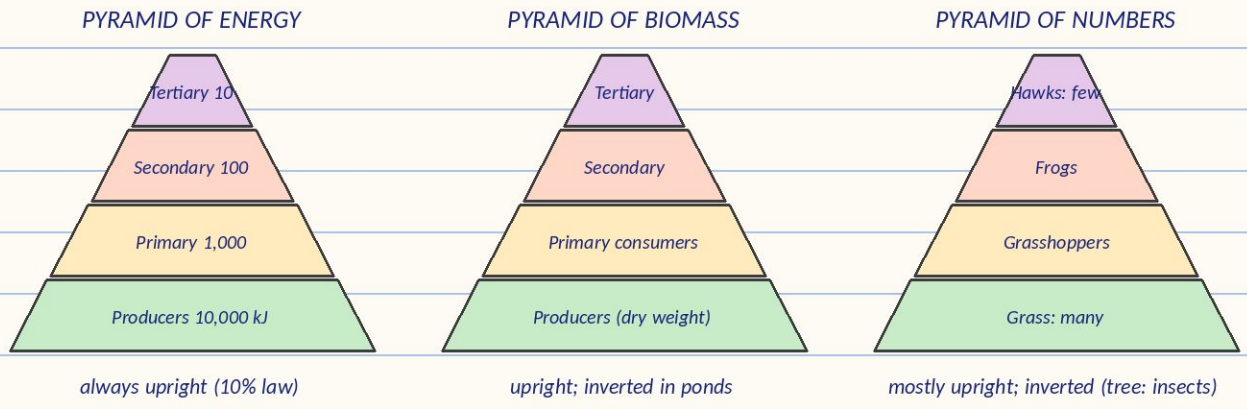


Fig: Ecological pyramids

- *Pyramid of energy always upright; biomass decreases upward; numbers usually decrease.*

Q3. [SLO 6, 7] Describe the greenhouse effect. Give the consequences of an increase in greenhouse gases.

Ans:

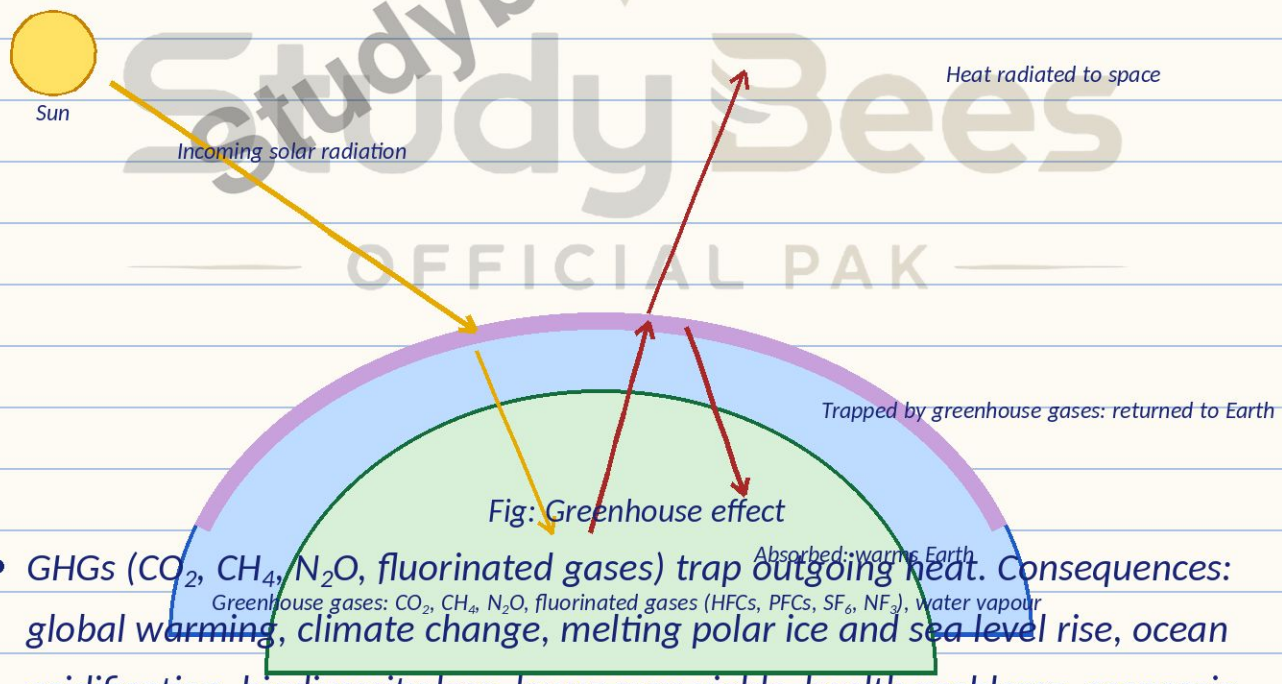


Fig: Greenhouse effect

- *GHGs (CO_2, CH_4, N_2O, fluorinated gases) trap outgoing heat. Consequences: global warming, climate change, melting polar ice and sea level rise, ocean acidification, biodiversity loss, lower crop yields, health problems, economic losses.*

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Q4. [SLO 8] Give the significance of coral reefs. Explain the effects of ocean acidification on coral reef ecosystems.

Ans:

- *Significance: hotspots of biodiversity, shelter and food for thousands of species, natural barriers protecting coasts, symbiosis with algae.*

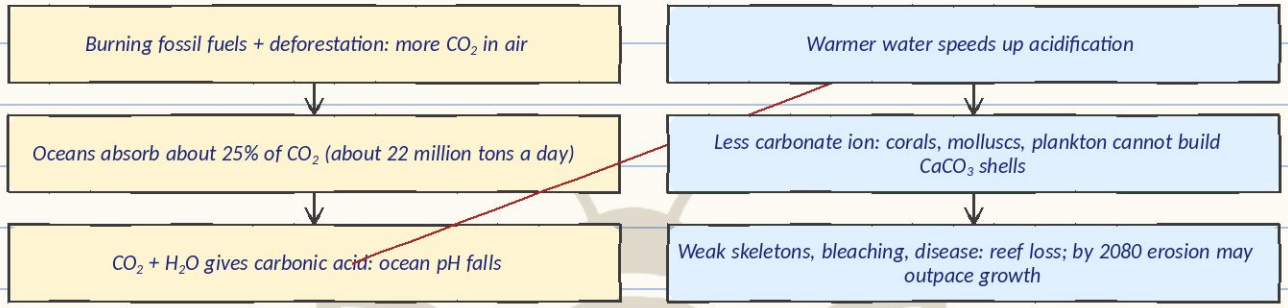


Fig: Ocean acidification and coral reefs

- *Acidification weakens skeletons, raises disease and predation, threatens more than half of corals; by 2080 erosion may outpace growth.*

Q5. [SLO 9-11] Define biogeochemical cycles and locate the reservoirs; describe the water cycle in detail.

Ans:

- *Reservoirs: living organisms and detritus; fossilized organic matter; dissolved inorganic nutrients; rocks.*

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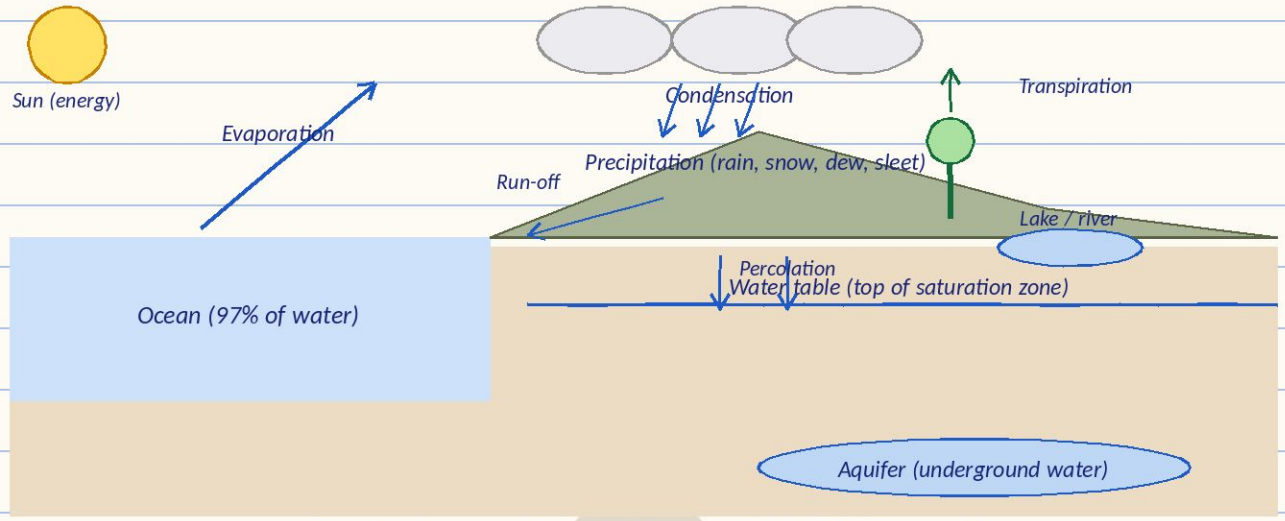


Fig: Water cycle

- *Water cycle: evaporation by solar energy, condensation, precipitation by gravity, run-off, percolation to water table and aquifers, transpiration, respiration.*

Q6. [SLO 12] Discuss the nitrogen cycle in detail.

Ans:

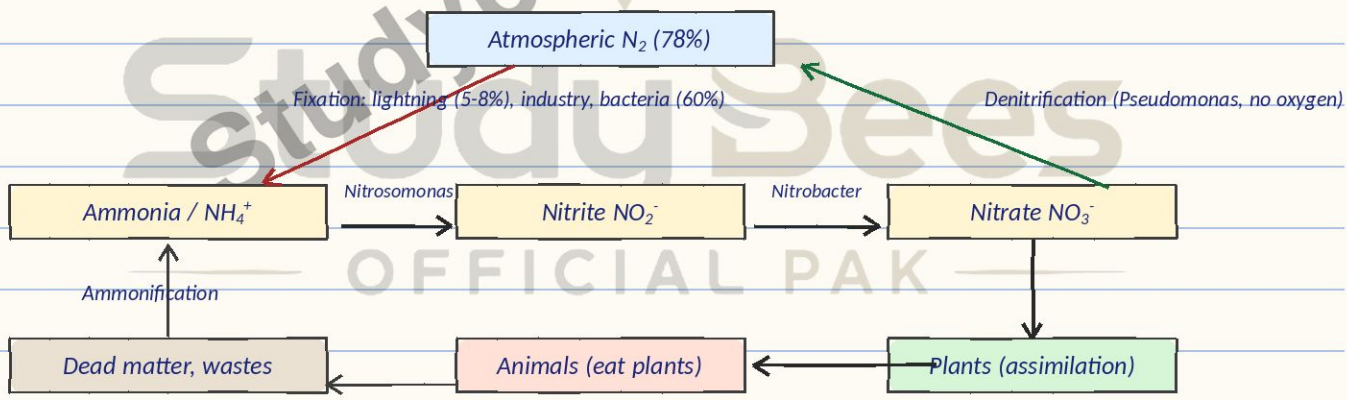


Fig: Nitrogen cycle

- *Fixation (atmospheric, industrial, biological), ammonification, nitrification (Nitrosomonas, Nitrobacter), assimilation, denitrification (Pseudomonas).*

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Q7. [SLO 17] Describe productivity in terms of GPP and NPP.

Ans:

- GPP: total solar energy fixed by producers. NPP = GPP - respiration; it appears as biomass and supports the food web.

Q8. [SLO 13-15] What is succession? Give the stages of primary succession on a barren rock.

Ans:

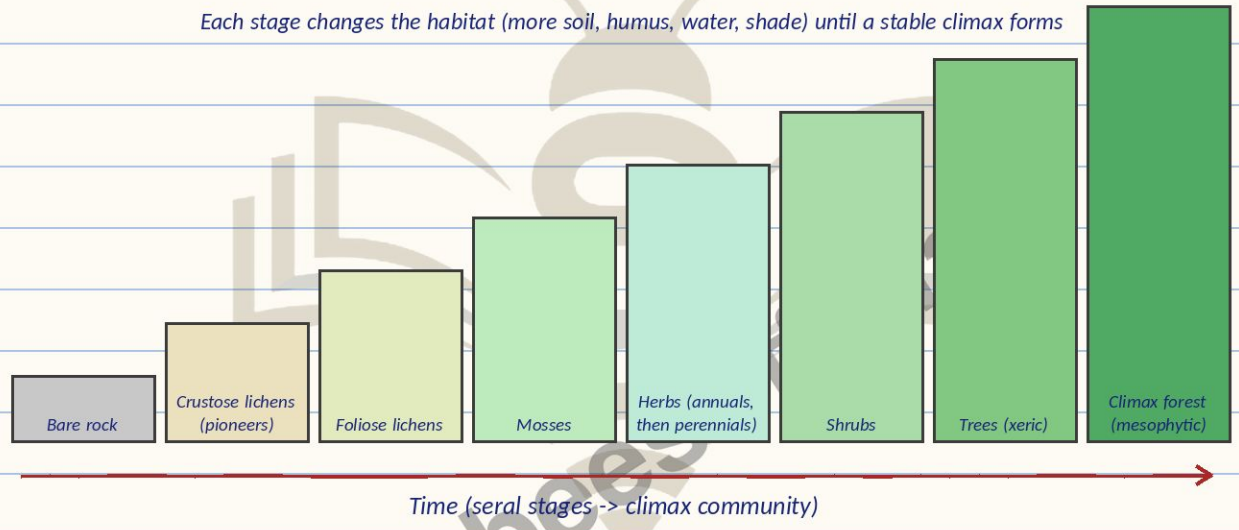


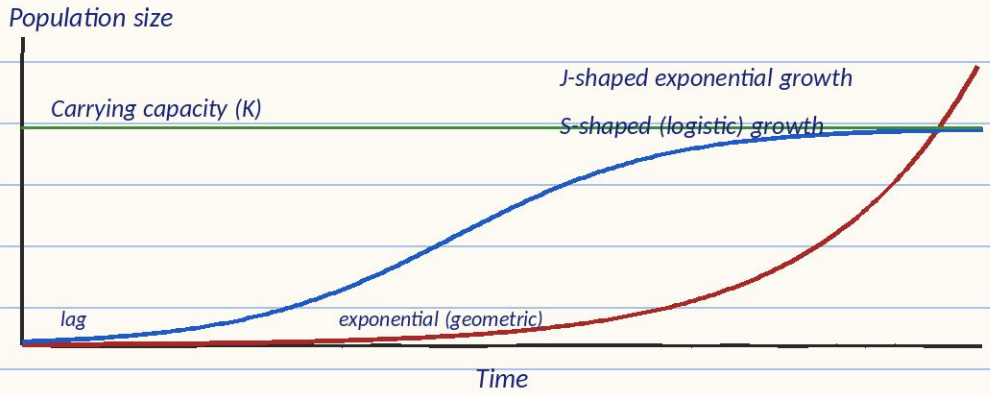
Fig: Xerarch (primary) succession on bare rock

- Succession: replacement of species until a stable climax. Primary succession (xerarch) on bare rock: crustose lichen, foliose lichen, moss, herb, shrub, tree, climax mesophytic forest. Secondary succession starts where soil exists; hydrarch starts in water.

Q9. [SLO 18, 19] Describe characteristics of a population and the effect of human population growth on ecosystems.

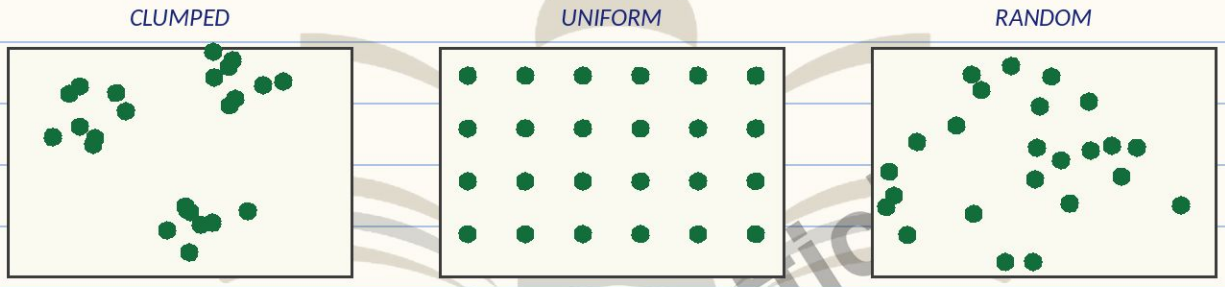
Ans:

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Lag phase, exponential phase (doubles at regular intervals), then growth levels off at K

Fig: Population growth curves



Clumped: patches (resources unevenly spread); Uniform: even spacing (competition); Random: unpredictable

Fig: Distribution patterns in a population

- Growth, density, distribution, carrying capacity, minimum viable size. Human impact: habitat destruction, pollution, climate change, overexploitation, invasive species, land use change, biodiversity loss, resource depletion.

Q10. [SLO 21] Describe grassland ecosystems in Pakistan.

Ans:

- Tropical grassland (savanna): monsoon; 200-600 mm rain; summer above 40 C; moderate to high productivity. Temperate grassland: 150-500 mm rain; Balochistan, Punjab; lower productivity.
- Flora: Cenchrus ciliaris, Themeda anathera, Apluda mutica, Panicum antidotale. Fauna: deer, antelope, gazelle, wild goat, rodents, birds; predators fox, wolf, jackal, big cats. Threats: overgrazing, erosion, urbanization; conserve by sustainable management.

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Q11. [SLO 21] Describe the desert, coniferous and temperate deciduous forest ecosystems of Pakistan.

Ans:

- *Desert: Thar, Tholistan, Thal, Kharan, Skardu cold desert; 100-250 mm rain; xerophytes; low productivity; adaptations of animals.*
- *Coniferous alpine and boreal forests: Western Himalaya and Karakoram; snow; pine, spruce, fir, cedar; snow leopard, brown bear; slow productivity; hotspots.*
- *Temperate deciduous forests: northern regions; oak, maple, walnut, deodar; birds, bears, deer, monkeys; threatened by deforestation.*

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